

A SIG on Understanding and Overcoming Barriers in Establishing HCI Degree Programs in Asia

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ABSTRACT

Despite a high demand for HCI education, Asia's academic landscape has a limited number of dedicated HCI programs. This situation leads to a brain drain and impedes the creation of regional HCI centers of excellence and local HCI knowledge. This SIG aims to gather stakeholders related to this problem to clarify and articulate its facets and explore potential solutions. The discussions and insights gained from this SIG will provide valuable input for the Asia SIGCHI Committee and other organizations in their endeavors to promote and expand HCI education across Asia. Furthermore, the findings and strategies identified can also serve as valuable insights for other communities in the Global South facing similar challenges. By fostering a comprehensive understanding of the barriers and brainstorming effective mitigating strategies, this SIG aims to catalyze the growth of HCI programs in Asia and beyond.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI);

KEYWORDS

HCI education, degree programs, Asia

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1 INTRODUCTION

The ACM SIGCHI has long promoted Human-Computer Interaction (HCI) education through projects such as the ACM SIGCHI Curricula for HCI [4], the SIGCHI Education Project [2], workshops, and the EduCHI symposia [3, 8, 9, 11, 16–18]. Yet, opportunities for HCI education remain concentrated in the Western world. A survey published in 2022 [12] identified 34 HCI Master's degree

programs across the globe. Out of the 34 identified programs, 31 are in the United States (12), Europe (17), or Australia (2). In Asia, there are only three Master's programs in HCI.

In contrast to the lack of programs, the demand for HCI education is reported to be high in many Asian countries [5, 7, 15]. Given this discrepancy, many students meet their educational needs by taking courses offered online by Western academics¹ or corporations². While these courses can provide high-quality practical knowledge and skills for individual students [10], factors such as their notoriously high dropout rates [14] suggest they are not suitable for all learners. Furthermore, a degree program can provide a richer scholarly environment for both students and faculty members. Developing such environments is essential for creating situated, local HCI knowledge instead of only importing concepts and methods from the West [15]. This also aligns with the principles of 'postcolonial computing' [6], which calls for a more inclusive, context-sensitive approach to HCI that respects and integrates diverse cultural experiences and knowledge systems, thereby contributing to a more equitable global HCI landscape. The availability of opportunities to continue HCI education locally may also reduce the brain drain: students emigrate first to study and, after graduation, continue their work career outside their home country. This practice, and the lack of local expertise it implies and engenders, may be impeding the development of a robust community and industry of HCI practitioners in Asia.

So, despite the importance of HCI degree programs, why are there so few in Asia? Sari and Wadhwa propose an explanation based on a summary of a workshop on HCI education at OzCHI 2014. They pointed out that universities, governments, and societies in Asia have yet to recognize the importance of HCI education [13]. However, beyond this hypothesis, we are not aware of any further efforts to articulate this problem. We believe that it is crucial to enhance our understanding of this problem and explore potential solution spaces for two reasons. Firstly, it is beyond the power of individual HCI educators to resolve this situation independently, and secondly, there may be opportunities for established organizations such as SIGCHI to intervene and promote the development of HCI degree programs in Asia. For these reasons, we propose this Special Interest Group meeting at CHI 2024. Despite the possibility of presenting as a dedicated session at the EduCHI symposium being held at CHI 2024, we believe a SIG meeting would be a more suitable venue to discuss these issues because of both its tight focus

*On behalf of the Asia SIGCHI Committee <https://sigchi.org/people/committees/asia-committee/>.

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¹e.g., Interaction Design Specialization from the University of California San Diego: <https://www.coursera.org/specializations/interaction-design>

²e.g., Google UX Professional certificate: <https://www.coursera.org/google-certificates/ux-design-certificate>

and the reduced commitment it entails. This will make it better able to attract important stakeholders and participant groups who are unlikely to be able to attend a prolonged event in the pre-conference period (e.g., Asian students who study abroad, employers/industry).

The SIG meeting will be organized as a hybrid synchronous meeting during CHI 2024 to foster an environment of collaborative discussion and knowledge sharing. The session will commence with a brief presentation providing an overview of the key challenges in establishing HCI degree programs in Asia. After setting the stage, attendees will be divided into focused breakout groups, each facilitated by a senior researcher well-acquainted with the topic. These groups will engage in in-depth discussions, exploring the identified challenges and brainstorming potential solutions. To conclude the SIG, a brief plenary summary session will be held where representatives from each group will share their key takeaways. This structure ensures a balanced blend of informative presentation, active engagement, and shared learning, ultimately aiming to generate a comprehensive understanding of the barriers and potential strategies to establish HCI degree programs in Asia.

The Asia SIGCHI Committee (ASC) was established in 2022 as an arm of the SIGCHI EC to increase the capabilities of HCI research and education in Asia. The ASC consists of 13 members from various Asian countries³. One of the missions of the ASC is to identify issues related to *HCI education in Asia*; it has determined the topic introduced in this SIG proposal as a key priority. As such, the authors of this proposal received a mandate from the ASC to lead this effort and will be fully supported by the ASC as they organize this SIG.

2 TOPICS FOR DISCUSSION

There is a broad range of open questions regarding challenges and barriers in establishing HCI degree programs in Asian countries. This SIG aims to identify the most pressing issues for HCI education across Asia, explore the underlying causes, and identify the next steps that can move the Asian HCI community forward towards a future in which it supports a more diversified, coherent, robust, and locally situated eco-system for fundamental HCI education. The SIG will target a diverse range of groups, including faculty members who are interested in establishing HCI courses and programs in Asia, Asian and non-Asian students who want to study HCI in Asia, students who want to study HCI in Asia but have not found appropriate programs or viable opportunities, and practitioners and researchers from industries in Asia.

To facilitate the discussion among the attendees, we will especially focus on the following themes, each of which we exemplify through a series of open questions:

- **The need for HCI programs in Asia:** Is a degree program for HCI in Asia necessary in light of many shorter “certificate” programs that are available internationally (e.g., Google UX certificate)? How could HCI degree programs be positioned so they are distinct from these certificates? What could HCI programs in Asia achieve? How would HCI in Asia change (e.g., in terms of jobs, student enrolment levels, etc.) if programs were to open? How would the proliferation of HCI degree programs in Asia change the global HCI

communities? What do students need from HCI programs in Asia? What does industry in Asia need from HCI graduates? Should HCI programs in Asia focus more on practical skills (e.g., UX or prototyping) or research skills?

- **Barriers in establishing HCI programs in Asia:** What are current barriers at the institutional level (i.e., universities, funding agencies, governmental agencies in higher education) that prevent the foundation of HCI degree programs in Asia? Do institutions in Asia simply lack the critical mass of faculty to offer HCI programs? If so, why? Is it enough to have HCI as individual courses inside Computer Science in Asia? What are the weaknesses of this setting? Would a USA-style model (e.g., with prominent industry engagement and sponsorship) work in the Asia context? How could sponsors be attracted and retained? What regional issues prevent the establishment of HCI programs in Asia, and how similar or different are these barriers across Asia?
- **Consequences of the status quo:** Does the lack of HCI programs in Asia contribute to the lack of HCI-related jobs in the area? What perspectives do Asian students who study HCI in Western countries have regarding this situation? How should we address the issues of the lack of funding opportunities for HCI-related research and teaching in Asia? What would be the lasting impacts of the lack of HCI degree programs in Asia in the long run, both to Asia and to the broader HCI community?
- **Broader collaborations and outreach:** How should we support the creation of HCI programs in Asia that require interdisciplinary courses from different schools/departments? How should we better leverage the existing infrastructure to foster the establishment of HCI programs, considering that in some universities, there might be many courses related to HCI being offered by different departments? How do we create a better eco-system for the students to learn HCI, given that there might be many HCI professors conducting research and teaching HCI-related courses across different departments/schools in Asia? How can we leverage the “new normal” (skills and infrastructure to study and collaborate remotely) in order to accelerate comprehensive HCI programs in Asia? What are the barriers to such settings? What can SIGCHI do to help? What lessons can we learn from other countries in the Global South that are establishing HCI programs (e.g., Brazil [1])?

3 GOALS AND EXPECTED OUTCOME

The primary objectives of this SIG are twofold. First, we aim to elucidate and pinpoint the key challenges inhibiting the establishment of HCI degree programs in Asia. Our goal is to foster a comprehensive understanding of the unique cultural, social, economic, and educational factors in Asia that may act as barriers to the development of HCI programs. This includes investigating the role of governmental policies, institutional support, curriculum development, and the availability of trained faculty, among other factors. Second, we aim to determine effective strategies to address these identified

³<https://sigchi.org/people/committees/asia-committee/>

barriers. This involves brainstorming innovative solutions, sharing best practices from successful HCI programs globally, and discussing potential collaborations and partnerships that can aid in overcoming these barriers. We will also explore how technological advancements can be leveraged to facilitate the establishment of HCI programs in Asia.

We expect to continue the discourse about the issues of this SIG through future ASC events that will unfold throughout 2024. Furthermore, we aim to organize a subsequent event at CHI 2025—which will take place in Asia—to continue the conversation with a likelihood to include more participants in Asia.

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